Core Features

1. Create and manage FTD users
2. Allow FTD users to login and authenticate and use the FTD app
3. Use websites, social media and/or documents to create and manage stock information for FTD app
4. FTD users will view different FTD web pages, FTD capabilities and FTD datasets to gain information about stocks or stock groups on the FTD app that the FTD user is interested in
5. FTD users will be able to leverage the FTD app to align or associate company stock information to company stock tickers and stock values
6. The FTD app will constantly have new stock ticker, new company stock information, and any other related information to the stock or company being pumped into this app, and then be made usable by the FTD app for the FTD user
    a. Think data pipelines to ingest different types of data and make that data usable for an application
    b. Considering using some type of base LLM with RAG
7. FTD users will be able to search stocks of interest in the FTD app
8. FTD users can click directly from summarized claims to the original primary source documents that the FTD app retrieved (SEC filings, whitepapers, press releases) to verify information integrity and perform independent due diligence
9. FTD users can group multiple companies, tickers, or technology topics into custom workspaces, serving as a personalized command center to consolidate and track ongoing research across fragmented sources.
10. FTD users can view a chronological mapping of key corporate events (e.g., product launches, earnings calls, regulatory filings) alongside associated news narratives, allowing them to visualize how specific current events have shaped the conversation surrounding a stock during its lifetime.

Core Requirements

ID	Description	User Action / Trigger	Acceptance Criteria (Given, when, then)	Non-Functional Requirements
REQ-001	User account creation	User submits a registration form.	Given a new user, when they submit valid credentials, then the system must create an encrypted profile.	Data must be stored with AES-256 encryption.

REQ-002	User authentication	User enters credentials at login.	Given a registered user, when they provide correct credentials, then they must be granted an active session token.	Session must timeout after 30 minutes of inactivity.
REQ-003	Data ingestion management	System pulls data from external sources.	Given a new source URL, when the ingest job runs, then the system must parse and store metadata in the database.	Pipeline must handle at least 500 concurrent source updates.
REQ-004	Information dashboard display	User navigates to a stock dashboard.	Given an authenticated user, when they select a ticker, then the UI must display relevant data within < 3 seconds.	Frontend must be responsive (mobile/desktop).
REQ-005	Ticker/Value mapping	System links metadata to ticker IDs.	Given raw company data, when mapped to a ticker, then the system must associate attributes to that specific stock.	System must implement a confidence-threshold filter (e.g., discard links with <70% confidence)

REQ-006	Data pipeline processing	System processes raw feed data.	Given raw incoming data, when ingested, then the RAG pipeline must generate summarized content for the user.	LLM latency must not exceed 10 seconds per summary.
REQ-007	Global search functionality	User inputs search query.	Given a search term, when the user triggers the search, then the system must return relevant stock entities.	Search results must be delivered within < 5 seconds.
REQ-008	Audit trail linking	User clicks on an AI-generated claim.	Given a summarized claim, when the user clicks the citation, then the system must navigate them to the primary source.	Links must be validated for 404/broken status at retrieval and every 24 hours.
REQ-009	Workspace management	User creates/edits a custom workspace.	Given a user's account, when they add a ticker to a workspace, then the system must persist this association to their profile.	Storage must allow up to 50 tickers per workspace.

REQ-010	Narrative timeline visualization	User views the event timeline.	Given a stock entity, when the user views the timeline, then the system must display events in chronological order.	Events must be indexed by timestamp from the ingested data.